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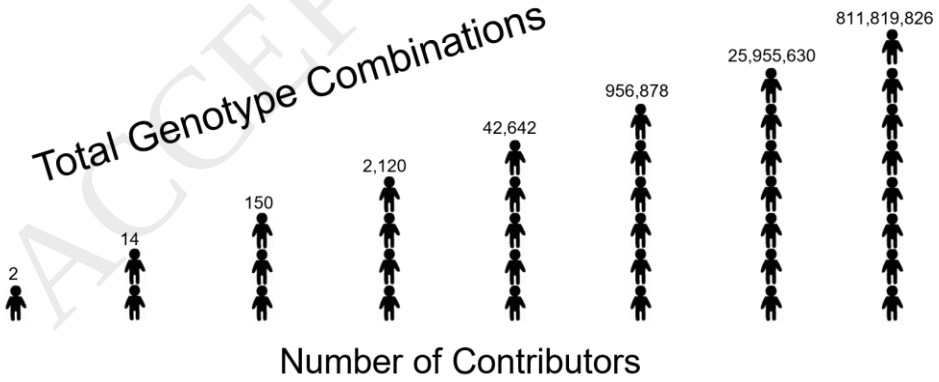
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Graphical abstract



Highlights:

- Deconvolution of the contributing genotypes in DNA mixtures containing greater than two contributors becomes increasingly difficult as the number of contributors increases.

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- Calculation, and enumeration, of the exact number of genotype combinations that are possible in a mixture containing four to eight contributors has not been done.
- The number of possible genotype combinations which can occur in DNA mixtures of two to eight contributors has been calculated using the concept of ordered partitions.
- The exponential increase in the number of possible genotype combinations which can make up mixtures of two to eight contributors underscores the difficulty in attempting analysis of mixtures without use of computer assisted analysis software.

Abstract

When DNA profiles contain a mixture of contributors (N), the maximum number of alleles observed at any locus is 2N with the number of alleles varying between 1 and 2N. The number of possible combinations of the genotypes of the N contributors increases as N increases. The number of possible combinations of genotypes that may occur among the N contributors can be enumerated using the mathematical concept of ordered partitions. Thus the number of possible genotype combinations which can exist at an STR locus in a DNA sample having N contributors and L observed alleles at the locus can be calculated. Development of the calculation and the resulting number of possible combinations which exist for N ≤ 8 contributors is presented. The results for ≥ three contributors indicate that deconvolution of contributing genotypes using manual methods is precluded except in circumstances where a clear major contributor can identified.

Keywords: DNA mixtures, number of contributing genotype possibilities, mixture deconvolution

Introduction

The number of contributors to a mixture that can be detected using current STR analysis kits and analysis methods may be as high as six to eight. Evidence samples containing DNA mixtures with three or more contributors are not uncommon. Given sufficient DNA in a mixture, the number of contributors detected will be limited by the probability of identity of the most discriminating loci of the test group [1]. Analysis of the resulting DNA profiles and elucidation with accompanying probabilities of possible contributor genotypes is difficult or impossible without the use of software designed specifically for that purpose [2–5]. Software used for DNA mixture analysis and comparison incorporates discrete continuous or qualitative continuous methods to model allele drop out. Modeling of other profile variables such as peak height (PH), peak height ratio (PHR), % stutter, DNA degradation and contributor ratio can be incorporated [3–7]. Most of these models require the user to assume a specific number of contributors (NOC). This number will usually be based on the maximum allele (MAC) count observed at one or more loci within the profile, where $NOC = MAC/2$. While this calculation reliably provides the lower bound of the possible NOC, the actual NOC may be higher. Recently other approaches to determining the NOC which incorporate modeling of the profile have been published and shown to provide a significant improvement in the estimate of the NOC [8–10].

While calculation of a likelihood ratio (LR) is the recommended approach for providing a quantitative measure of the weight of the evidence, some laboratories have calculated the probability of a random person being included as a contributor to an observed DNA mixture using the combined probability of inclusion/combined probability of exclusion (CPI/CPE), when DNA results include a known individual as a possible contributor to a mixture based on observation of the known person's alleles in the mixture [11–13]. Typically, use of this method requires validation and definition of an analytical threshold and a stochastic threshold [14,15]. Appropriate use of the CPI requires that sufficient DNA has been amplified such that all alleles are observed (i.e. there is no allele drop out). The preferred alternative to the CPI is calculation of a likelihood ratio which includes use of a value or modeling of values, for allele drop out [2–6].

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Two person mixtures where all alleles are present can sometimes be de-convoluted "by hand" depending on the specific genotype combinations present and when the ratio of contributors allows for designation of major and minor contributors. De-convolution may also be possible when the DNA has come from a vaginal swab or other type of intimate sample such that one individual can be assumed to be present and that individuals' profile is known. In both circumstances allele sharing may still result in some ambiguity. In a two person mixture where all alleles are present, it is possible to visualize the 14 possible combinations of the two genotypes which result in observations of 1, 2, 3 or 4 alleles at each locus and to consider which genotype combinations "make sense". This process uses/accounts for other profile data such as peak heights, peak height ratios, contributor proportions and other information from the mixture.

Overson and Butler have presented charts showing the genotype combinations which can be found in two and three person mixtures [16,17]. (For three person mixtures see DNA Box 7.2, page 167, and Figure 7.4, page 168, in Chapter 7 *Low-Level DNA and Complex Mixtures* from reference 18.) Table 1 below gives the number of different combinations of genotypes which produce from 1 to 6 alleles in a three person mixture. The total number of possible combinations is 150. These 150 combinations can be subdivided by the number of alleles observed at a locus and can be further subdivided by "classes" of combinations characterized by the homozygous/heterozygous states of each contributor and the number of alleles and genotypes shared between contributors.

Table 1: Distribution of the number of possible combinations of three contributors which will have from one to six alleles at a locus [16,17].

Number of alleles observed (L)	1	2	3	4	5	6
Number of possible genotype combinations [G(L)]	1	8	29	52	45	15

Considering that a locus can have anywhere between 1 and 6 alleles, deciphering the genotypes of the three contributors requires unraveling which of the possible combinations of genotypes for the number of alleles observed, is actually present for each locus. Thinking about the possibility of doing this should quash any thoughts that the contributor genotypes can be deduced by hand even when considering the observed peak heights. The number of possible genotype combinations G(L) that exist in mixtures having more than three persons has not been calculated but will be greater than 150. In this paper we derive a method for computing the number of possible allocations of genotypes among N contributors of the genotypes derived from a set of alleles at a single locus. We present a definition of "class" that allows separation of the genotype combinations having a given total number of alleles (L) into classes which are consistent with those shown by Overson and Butler. The definition is expandable to any number of contributors. While knowing the number of possible genotype combinations that would account for the presence of, *e.g.* 1 to 8 alleles at each locus of a four person mixture, does not contribute to solving the problem of determining the likely genotypes of the four contributors, it illustrates the magnitude of difficulty of the problem. This information also illustrates why including an individual solely because his/her alleles are present is hazardous.

Computation of Possible Genotype Allocations

Defining $\mathcal{G}(\mathcal{L})$

Suppose we have observed a DNA profile from multiple contributors which has a set $\mathcal{L}$ of alleles at a particular locus, and suppose $\mathcal{L}$ has L elements (*i.e.*, L alleles are observed). Let $\mathcal{G}(\mathcal{L})$ be the set of possible contributing genotypes that can be formed by the alleles of $\mathcal{L}$. The number of elements in $\mathcal{G}(\mathcal{L})$ is

$$G(L) = L + L^* (L-1)/2 = \tfrac{1}{2}(L^2+L)$$

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where L of the genotypes are homozygous, and L*(L-1)/2 are heterozygous.

For example

For L=4 alleles, labeled A_1, A_2, A_3, A_4, there are G(4)= ½(4^2+4)=10 genotypes which could contribute to the observation of A_1, A_2, A_3, A_4 at the locus: These possible contributing genotypes are:

(A_1,A_1), (A_2,A_2), (A_3,A_3), (A_4,A_4), (A_1, A_2), (A_1,A_3), (A_1,A_4), (A_2,A_3), (A_2, A_4), (A_3, A_4).

Defining $\mathcal{D}(N,\mathcal{L})$

Let $\mathcal{D}(N,\mathcal{L})$ be the set of possible allocations of the genotypes in $\mathcal{G}(\mathcal{L})$ from N contributors to a mixture in which each allele from $\mathcal{L}$ is present in at least one contributor. Two allocations are considered to be identical if they differ only by a permutation of genotypes among the contributors. (We emphasize here that our "contributors" should be thought of as dummy placeholders for genotypes, and not as actual contributors in a forensic sense. Thus, a permutation of genotypes among contributors, such as swapping genotypes (A_1,A_1) and (A_2,A_2) between two contributors, is essentially meaningless for purposes of our computation.)

Because all alleles observed in the mixture must be present in each allocation to N contributors and because alleles may be shared by 2 or >2 contributors, it follows that $L \le 2 \cdot N$. Let D(N,L) be the number of elements in $\mathcal{D}(N,\mathcal{L})$. We seek to compute D(N,L).

Defining $\mathcal{U}(N,\mathcal{L})$

Initially, to simplify the computation, we lift the requirement that all alleles from $\mathcal{L}$ be present. Later in the computation we will discard those allocations in which one or more alleles are missing.

Let $\mathcal{U}(N,\mathcal{L})$ be the set of all possible allocations of the genotypes in $\mathcal{G}(\mathcal{L})$ among N contributors, where we allow the possibility that not all alleles are present. As for $\mathcal{D}(N,\mathcal{L})$, two allocations are also considered identical if they differ only by a permutation of genotypes among the contributors. Let U(N,L) be the number of elements in $\mathcal{U}(N,\mathcal{L})$.

To compute U(N,L), we first impose an ordering on the genotypes. The alleles are identified as A_1, A_2, A_3, …, A_L. Each genotype is of the form (A_i, A_j), and we assume $i \le j$. To each genotype (A_i, A_j) we assign the integer $L \cdot (i-1)+j$. If G is the genotype given by (A_i, A_j), we assign a number O(G) to the genotype G by:

$$O(G) = O(A_i, A_j) = L \cdot (i-1)+j.$$

This assigns a unique integer to each genotype and allows us to order the genotypes by increasing O-values. Note that since $i \le j$, not all possible $L \cdot (i-1)+j$ values appear. This does not matter; the point is simply to be able to list the genotypes in some order.

Example:

For 4 alleles, A1, A_2, A_3, A_4, the O-values of the 10 genotypes are:

$O(A_1,A_1) = 1$, $O(A_1,A_2)=2$, $O(A_1,A_3)=3$, $O(A_1,A_4)=4$,
$O(A_2,A_2)=6$, $O(A_2,A_3)=7$, $O(A_2,A_4)=8$,
$O(A_3,A_3)=11$, $O(A_3,A_4)=12$,
$O(A_4,A_4)=16$

Note that all we're doing here is assigning an O-value as if the (A_i, A_j) genotype were the (i,j) entry of a matrix

Given an allocation G_1, G_2, G_3, …, G_N of genotypes among the N contributors we may assume that

$$O(G_1) \le O(G_2) \le O(G_3) \le \ldots \le O(G_N)$$

We must consider the case where several contributors have the same genotype. For example it may be that G_1, the first genotype in the allocation, is the same as G_2.

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In the general case, we might have something like

$G_1, \ldots, G_r, G_{r+1}, \ldots, G_{r+s}, \ldots \ldots, G_{N-t}, \ldots, G_N$
$\backslash \ldots\ldots / \quad \backslash \ldots\ldots\ldots / \qquad\qquad \backslash \ldots\ldots\ldots / \qquad$
$\quad | \qquad\qquad | \qquad\qquad\qquad | \qquad$
$\quad |$ s identical genotypes $\qquad$ t identical genotypes
r identical genotypes

In order to handle these "runs" of identical genotypes in our calculation we introduce the concept of an *ordered partition*.

Definition: An *ordered partition* of a positive integer N is a sequence of positive integers $(r_1, r_2, r_3, \ldots, r_K)$ such that $r_1+r_2+r_3+\ldots+r_K = N$. The order matters here: (1,2) and (2,1) are *different* ordered partitions of 3. Ordered partitions are sometimes called *combinations* in the mathematical literature [18].

Examples:
There are 4 ordered partitions of 3: (3), (1,2), (2,1) and (1,1,1).
There are 8 ordered partitions of 4: (4), (1,3), (3,1), (2,2), (1,1,2), (1,2,1), (2,1,1), and (1,1,1,1).

Given an ordered partition $(r_1, r_2, r_3, \ldots, r_K)$ of N, choose K *distinct* genotypes $G_1, G_2, G_3, \ldots, G_K$ with the property that $O(G_1) < O(G_2) < O(G_3) < \ldots < O(G_K)$. Take r_1 copies of G_1, r_2 copies of G_2, . . ., r_K copies of G_K. This gives an allocation $G_1, \ldots, G_1, G_2, \ldots, G_2, \ldots \ldots, G_K, \ldots, G_K$ in which each genotype in the sequence has an O-value strictly less than its successor.

Example:
The ordered partition (3,1) gives the allocation G_1, G_1, G_1, G_2.
The ordered partition (1,3) gives the allocation G_1, G_2, G_2, G_2.

Given an ordered partition of N with K elements, how many ways can we select K distinct genotypes from $\mathcal{G}(\mathcal{L})$? The answer is the binomial coefficient

$(G(L):K) = G(L)!/((K!)\cdot(G(L)-K)!)$

Example:
How many ways can we select 3 distinct genotypes from 10 possible genotypes?
Answer: $(10!)/(3!7!) = (10*9*8)/(3*2) = 120$.

So each ordered partition $(r_1, r_2, r_3, \ldots, r_K)$ of N gives rise to (G(L):K) distinct allocations of genotypes from $\mathcal{G}(\mathcal{L})$. For K=1, 2, . . ., N, let P(N,K) be the number of ordered partitions of N with K elements. See Appendix A for a means of computing P(N,K).

Example:
For N=4, P(4,1)=1, P(4,2)=3, P(4,3)=3, P(4,4)=1

Thus, given L alleles and N contributors, so $L \le 2\cdot N$, the total number of distinct allocations of the G(L) genotypes in $\mathcal{G}(\mathcal{L})$ is:

$$U(N,L) = \sum_{K=1}^{N} P(N,K)\cdot(G(L):K)$$

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Example:

Let N=4 and L=6. Then G(L)=21 and

$$U(4,6) = \sum_{K=1}^{4} P(4,K) \cdot (21:K)$$

U(4,6) = P(4,1)·(21:1) + P(4,2) ·(21:2) + P(4,3) ·(21:3) + P(4,4) ·(21:4)
U(4,6) = 1 · 21 + 3 · (21·20)/(2·1) + 3 · (21·20·19)/(3·2·1) + 1 · (21·20·19·18)/(4·3·2·1)
U(4,6) = 21 + 630 + 3990 + 5985
U(4,6) = 10626

Computation Of D(N,L)

$\mathcal{D}$(N,$\mathcal{L}$) consists of those allocations in $\mathcal{U}$(N,$\mathcal{L}$) from which no allele is missing, *i.e.*, all alleles are present. To remove from $\mathcal{U}$(N,$\mathcal{L}$) those allocations from which one or more alleles is missing, we proceed as follows: Consider the case where the first allele, A_1, is missing. There are D(N,L-1) allocations in $\mathcal{U}$(N,$\mathcal{L}$) that have all the alleles *except* A_1. Doing this for each of the L alleles, we conclude here are L· D(N,L-1) allocations in $\mathcal{U}$(N,$\mathcal{L}$) with exactly one allele missing. These must be removed from $\mathcal{U}$(N,$\mathcal{L}$). Consider allocations with exactly 2 alleles missing. There are D(N,L-2) allocations in $\mathcal{U}$(N,$\mathcal{L}$) with only A_1 and A_2 missing. There are (L:2) distinct pairs of alleles, so there are (L:2)· D(N,L-2) allocations in $\mathcal{U}$(N,$\mathcal{L}$) with exactly 2 alleles missing. These too must be removed from $\mathcal{U}$(N,$\mathcal{L}$). For 3 alleles, there are (L:3)·D(N,L-3) allocations in $\mathcal{U}$(N,$\mathcal{L}$) with 3 alleles missing. Continuing, we finally get to the number of allocations with all but one allele missing, *i.e.*, only one allele is present. There are (L:(L-1))· D(N,1) allocations with L-1 alleles missing. Subtracting these values for allocations with missing alleles gives

D(N,L) = U(N,L) – ((L:1)·D(N,L-1) + (L:2)·D(N,L-2) + (L:3)·D(N,L-3) + … + (L:L-1)·D(N,1))

or

$$\mathrm{D(N,L)} = \mathrm{U(N,L)} - \sum_{\mathrm{i=1}}^{\mathrm{L-1}} \mathrm{(L:i)} * \mathrm{D(N,L-i)}$$

Example:

$$D(4,6) = U(4,6) - \sum_{i=1}^{5} (6:i) * D(4,6-i)$$

D(4,6) = U(4,6) – ((6:1)·D(4,5) + (6:2)·D(4,4) + (6:3)·D(4,3) + (6:4)·D(4,2) + (6:5)·D(4,1)
Skipping the computations of D(4,1), D(4,2), …, D(4,5), which will have been done previously
D(4,6) = 10626 – (6·600 + ((6·5)/2)·297 + ((6·5·4)/(3·2))·84 + ((6·5)/2)·13 + 6·1)
= 10626 – (3600 + 4455 + 1680 + 195 + 6)
= 690

All Possible Combinations of N Contributors

All possible combinations of N contributors at a single locus is given by

D(N,1) + D(N,2) + . . . +D(N, 2·N)

Example:

All possible combinations of 4 contributors at a single locus is:

A(4) = D(4,1) + D(4,2) + D(4,3) + … +D(4,8)

Computing D(4,L) as above gives

A(4) = 1 + 13 + 84 + 297 + 600 + 690 + 420 + 105 = 2210

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The results of the computation of the possible number of genotype combinations which represent all possible ways that genotypes from N contributors can combine to produce up to 2N alleles is presented in Table 2 for N=1 through 8. These numbers increase dramatically as the number of contributors increases. For mixtures with more than 3 contributors the number of combinations is relevant as it clearly illustrates how difficult the deconvolution problem is and the necessity for computational/computer solutions versus manual solutions to the problem.

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Table 2: Computation of the possible number of genotype combinations which represent all possible ways that genotypes from N contributors can combine to produce ≤ 2N alleles

		Number of Alleles L																
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total
Number of Contributors N	1	1	1															2
	2	1	4	6	3													14
	3	1	8	29	52	45	15											150
	4	1	13	84	297	600	690	420	105									2,120
	5	1	19	192	1,116	3,933	8,661	11,970	10,080	4,725	945							41,642
	6	1	26	381	3,321	18,080	63,919	150,332	236,978	247,275	163,800	62,370	10,395					956,878
	7	1	34	687	8,484	66,645	346,644	1,231,857	3,052,008	5,316,885	6,483,330	5,415,795	2,952,180	945,945	135,135			25,955,630
	8	1	43	1,155	19,428	210,645	1,529,064	7,687,512	27,472,653	71,004,690	133,874,415	184,033,080	182,338,695	126,756,630	58,648,590	16,216,200	2,027,025	811,819,826

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Within each category defined by the number of contributors and the number of alleles present, different "classes" can be defined which correspond to the groupings of genotype combinations such as those shown in the chart in JM Butler Chapter 7, page 168 [17]. These subdivisions or "classes" can be defined as follows: Two genotype allocations are in the same class if and only if:

1. They have the same number of homozygous genotypes
2. They have the same number of heterozygous genotypes
3. The alleles (assuming a contributor ratio of 1:1:1:….:1, *i.e.*, all contributors to the sample contribute an equal mass) that appear in the two allocations are in the same proportions
4. The genotypes (again assuming a contributor ratio of 1:1:1:….:1) that appear in the two allocations are in the same proportions

The computation of the number of possible genotype combinations given a particular number of contributors that is developed in this paper does not directly lend itself to partitioning these combinations into classes as defined above. However, a computer program was written that generates the list of possible genotype combinations and sorts the list into classes as defined above. Along the way the program computes the proportions of alleles and genotypes that would be observed in the resulting mixture if each contributor contributed an equal mass of genotypes to the sample. Table 3 illustrates the class definitions for 1-6 alleles present when N=3. This table is consistent with the information presented in J.M. Butler (Chapter 7, page 168) and in Overson [16,17].

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Table 3: Class definition applied to three contributors (listing is not in same order as class listings in Butler and Overson)

# of alleles	*Class*	*# of Homozygous Contributors*	*# of Heterozygous Contributors*	*Genotype Ratio in 1:1:1 Mixture*	*Allele Proportion in 1:1:1 Mixture*
1	1.1	3	0	na	na
2	2.1	3	0	2:1	4:2
	2.2	2	1	2:1	5:1
	2.3	2	1	1:1:1	3:3
	2.4	1	2	2:1	4:2
	2.5	0	3	na	3:3
3	3.1	3	0	1:1:1	2:2:2
	3.2	2	1	2:1	4:1:1
	3.3	2	1	1:1:1	3:2:1
	3.4	1	2	1:1:1	4:1:1
	3.5	1	2	1:1:1	3:2:1
	3.6	1	2	2:1	2:2:2
	3.7	0	3	2:1	3:2:1
	3.8	0	3	1:1:1	2:2:2
4	4.1	2	1	1:1:1	2:2:1:1
	4.2	1	2	1:1:1	3:1:1:1
	4.3	1	2	1:1:1	2:2:1:1
	4.4	0	3	1:1:1	3:1:1:1
	4.5	0	3	2:1	2:2:1:1
	4.6	0	3	1:1:1	2:2:1:1
5	5.1	1	2	1:1:1	2:1:1:1:1
	5.2	0	3	1:1:1	2:1:1:1:1
6	6.1	0	3	1:1:1	1:1:1:1:1:1

It is important to note (assuming contributors are present in equal mass) that the resulting allele proportions are not unique to a class and occur in other classes having the same number of alleles. While a 1:1: 1:…:1 ratio of contributors may only rarely be seen, the assumption provides a simple way to compare the ratio of alleles within and between classes. Using this assumption provides a method to visualize how different classes, for a given L number of alleles, could have very similar allele ratios which, in the profile, are visualized as peak heights. These results again illustrate the complexity of the problem and illustrate that combinations of individuals, where different numbers of homozygous and heterozygous contributors are present, can result in profiles which have alleles present in the same ratios.

Discussion

The ideal DNA profile configuration for DNA mixture analysis has no allele drop out and has contributors present at different template mass amounts such that the height of the RFU signal can be used to distinguish the genotypes of the contributors including some or all of the genotypes of minor contributors. This optimal situation can occur in two and sometimes three person mixtures. However, complicating factors occur in DNA profile interpretation which can affect correct allele count. These

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include mis-identification of stutter as an allele (and the reverse) and allele drop out or drop in when low template amounts of DNA are present from one or multiple contributors [19,20]. Allele sharing between contributors confounds determination of the NOC using MAC [21–23]. In all profiles, cumulative process variation is observed as variation in percent stutter, peak height (PH), peak height ratio (PHR) and contributor ratio [24,25]. Characterization of these features of DNA profiles has assisted in the elucidation of the problems involved in profile interpretation and in the development of software solutions for DNA profile interpretation.

The data presented here characterize an additional dimension to the problem: Why are DNA mixtures difficult to interpret? Even in circumstances when all the alleles at each locus are present and have been correctly identified, the sheer number of possible genotype combinations, shown in Table 2, precludes deciphering of the genotypes of the individual contributors without computer/software assistance. Additionally, when considering the process of DNA profile interpretation, permutations of the assignment of genotypes to specific contributors and the effect of ambiguous peaks on the NOC results in the true process being complicated beyond the numbers of combinations presented here. Thus, the numbers of combinations presented in Table 2 represent the lower bounds given other complexities of DNA profile interpretation.

The data shown in Table 3 provides additional information to consider in planning validation of probabilistic genotyping methods. We show that mixtures can be created which will be similar in allele proportions but do not have the same set of contributor genotypes. Therefore, test mixtures can be created which may assist in defining the limitations of genotyping software. Most importantly, the data derived above define in greater detail the complexity of the problem of deconvolution of STR mixtures. These challenges are present and independent of short-comings of the PCR and detection processes.

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Appendix A

Computation of P(N,K)

For integers N and K with $N \geq K > 0$, let P(N,K) be the number of ordered partitions of N with K elements. Clearly, for all N>0, P(N,N) = 1 and P(N,1) = 1.

Given an ordered partition (R_1, . . ., R_k) of N with $k \geq 2$ elements, it follows that (R_2, . . ., R_k) is an ordered partition of N-R_1 with k-1 elements.

So for $N \geq 2$ and $N > K > 1$,

$$P(N,K) = P(N-1,K-1) + P(N-2,K-1) + \ldots + P(1,K-1), where\ P(r,k) = 0\ for\ r < k$$

$$= \sum_{r=1}^{N-1} P(r,K-1)\,, where\ P(r,k) = 0\ for\ r < k$$

Given that P(N,N) = 1 and P(N,1) = 1, we can use the above equation to compute P(N,K) recursively.

Example:

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$$
\begin{aligned}
P(5,3) &= P(1,2) + P(2,2) + P(3,2) + P(4,2) \\
&= 0 + (P(1,1)) + (P(1,1) + P(2,1)) + (P(1,1)+P(2,1)+P(3,1))) \\
&= 0 + 1 + 1 + 1 + 1 + 1 + 1 \\
&= 6
\end{aligned}
$$

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